

ROBOTICS PROJECT SHOWCASE

Hands-on Mobile Robot • Arduino UNO • DTMF Control • Motors • Gears
• Custom Fabrication



Project photograph — complete robot prototype

A practical robotics project integrating embedded electronics, audio-tone command input, electromechanical actuation, custom mechanical transmission, structural fabrication and audience interaction.

[Video Demonstration](#)



1. Project Photographic Evidence

The photographs show the physical prototype and demonstrate that the project involved a complete fabricated robotic system rather than a simulation-only implementation.

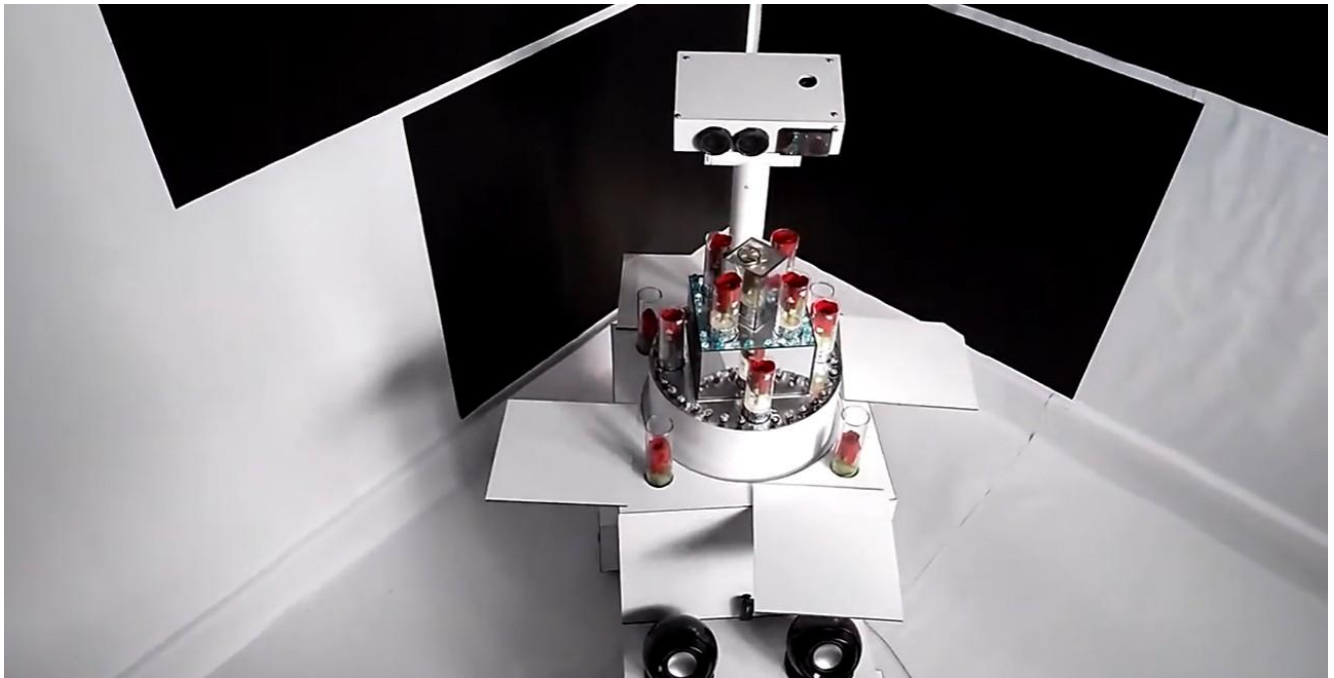


Figure 1. Close-up view of the robot's upper mechanism, rotating assembly and sensor/head structure.



Figure 2. Full prototype showing the custom body, wheel/motor area, vertical structure and upper mechanism.

2. System Overview

The robot was designed as an integrated mechatronic system. The Arduino UNO provides embedded control, while a DTMF decoder converts audio tones into digital commands. DC motors provide locomotion, servo motors operate mechanisms such as lids, and mechanical gearing converts and stabilizes motion.

Subsystem	Implementation
Controller	Arduino UNO
Command input	DTMF tones using a 16-pin DTMF decoder module
Locomotion	DC motors
Positioned mechanisms	Servo motors
Mechanical transmission	Rack gear, worm gear and threaded/gear mechanisms
Structure	Plywood and aluminium channel
Interaction	Bluetooth audio speaker for spoken communication / announcements

3. DTMF-Based Command and Control

A distinctive feature of the project is the use of DTMF (Dual-Tone Multi-Frequency) tones as the robot command interface. The DTMF module decodes the received tone into digital signals that can be read by the Arduino.

DTMF-Based Robot Control Logic



Key idea: audio tones are converted into discrete control commands.

The resulting control chain is: **audio tone • DTMF decoder • Arduino logic • actuator command • physical robot movement.**

4. Mechanical Motion and Transmission

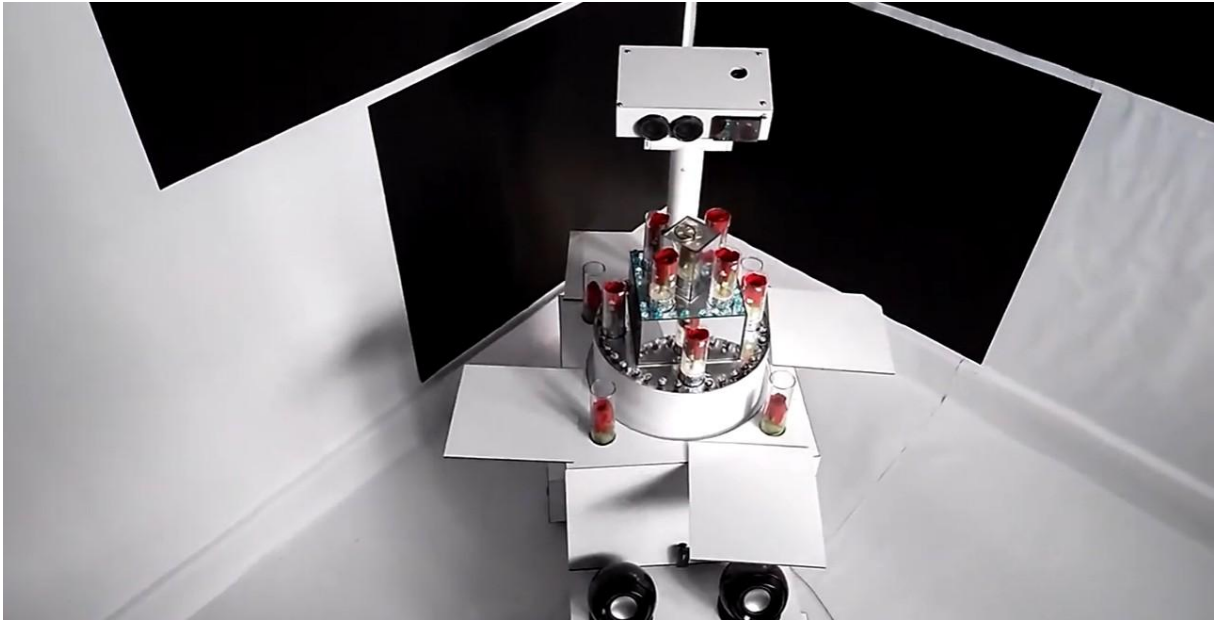
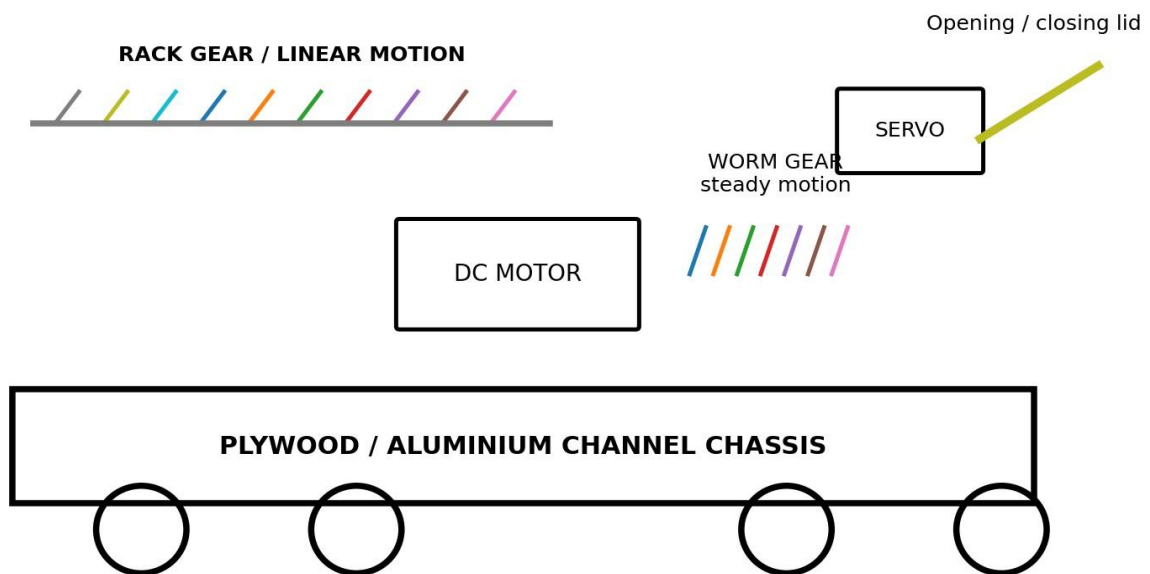


Figure 3. The upper rotating/mechanical assembly visible in the prototype.

Mechanical Construction & Actuation



Mechanical elements

- **DC motors:** provide the main robot movement.
- **Servo motors:** provide controlled angular motion for opening/closing lids and mechanisms.
- **Worm gear:** provides controlled, steady transmission and can help resist unwanted back-driving.
- **Rack gear:** converts rotational motion into linear movement.
- **Threaded/gear mechanisms:** support controlled mechanical movement and positioning.

5. Custom Fabrication and Structure



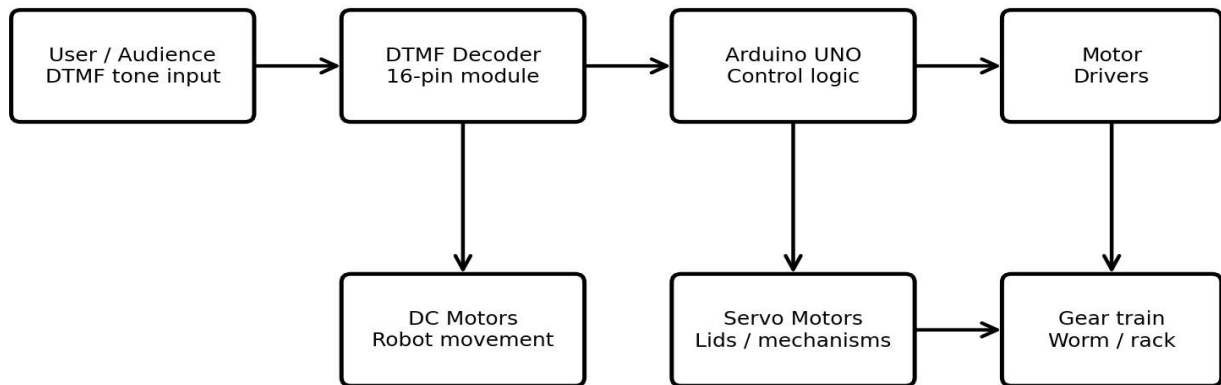
Figure 4. Full-body view of the custom-built robot.

The body was constructed using **plywood and aluminium channel**. This approach allowed the robot to be fabricated around the required electronics, motors and mechanisms. It also demonstrates practical skills in mechanical prototyping, mounting, alignment, wiring and system assembly.

Rather than using a ready-made commercial robotic platform, the project required integration of individually selected components into a working physical system.

6. Electronics, Embedded Software and Mechatronics

Control & Actuation Architecture



The project demonstrates the complete embedded control chain: **input • processing • actuation • mechanical response**.

Engineering area	Demonstrated capability
Embedded systems	Arduino-based real-time control
Electronics	DTMF decoder and actuator interfacing
Programming	Command interpretation and motor/servo control
Motor control	Coordinated DC motor movement
Servo control	Controlled mechanism positioning
Mechanical systems	Gear-based transmission and motion control
Mechatronics	Integration of software, electronics and mechanics

7. Human–Robot Interaction

A Bluetooth audio speaker was incorporated into the robot so that it could speak to or address the audience during demonstrations. This adds an interaction layer beyond movement and makes the robot suitable for live public demonstrations.

Interaction	Role
Robot movement	Provides visible physical response
DTMF commands	Provides a discrete command interface
Mechanical actions	Creates visible task-oriented behaviour
Bluetooth audio	Provides spoken communication / announcements
Audience engagement	Makes the demonstration interactive

8. Connection to Future Robot-Arm Research

This project provides a strong practical foundation for moving toward robot-arm research. The important progression is from discrete, pre-programmed commands toward systems that use sensing, perception, feedback and intelligent decision-making.

Previous project	Robot-arm research extension
Arduino embedded control	Robot-arm and gripper control
DTMF command interface	Human-robot task commands
DC motor experience	Actuator and motion-control concepts
Servo mechanisms	Joint and end-effector positioning
Rack/worm/threaded gears	Transmission, stability and precision
Custom fabrication	Experimental fixtures and robot-arm integration
Audience interaction	Human-robot interaction
Integrated electronics/software	IoT-connected and AI-assisted robotics

Suggested research transition: use the SO-101 robot arm as a platform for vision-based intelligent pick-and-place, then investigate adaptive feedback, learning from demonstration and IoT/cloud integration.

9. Short Academic Presentation Summary

“My previous robotics project was a hands-on mechatronics system in which I designed and built the physical robot and integrated the electronics, embedded software and mechanical transmission. I used an Arduino UNO as the controller, DTMF tones as a command interface, DC motors for locomotion and servo motors for controlled mechanisms. I also designed mechanical transmission using rack, worm and threaded/gear mechanisms and fabricated the structure using plywood and aluminium channel. A Bluetooth speaker provided an audience-interaction layer. This experience motivated me to progress from conventional embedded robot control toward research in robot arms, computer vision, intelligent manipulation and feedback-based control.”

This wording positions the project as evidence of **practical robotics, embedded systems, electronics, mechanical design, programming and mechatronics integration**—all relevant foundations for your new robot-arm research.